D. Hussaimiah
192425300

SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme: **B.E / B.Tech**
Course Code: **MLA02**
Course Name: **FUNDAMENTALS OF MACHINE LEARNING**
Duration: **1:30 Hr**

Date : 02/01/2026 Max Marks: **50 Marks** **ASSIGNMENT – I**

UNIT I – INTRODUCTION AND MATHEMATICAL FOUNDATIONS

Topics: *Design of a learning system, types of machine learning, probability theory and distributions, Bayes decision theory, information theory*

Course Outcome

On successful completion of this assignment, students will be able to:

CO	Course Outcome Statement
CO1	Analyse machine learning fundamentals and mathematical foundations, including probability theory, Bayes decision theory, and information theory.

Answer all the Tasks

Task 1. A retail bank plans five machine-learning systems. For each system, analyse the problem and state (i) the learning paradigm (supervised / unsupervised / semi-supervised / reinforcement), (ii) the learning task (classification, regression, clustering, control, etc.) and (iii) the experience E and performance measure P in Mitchell's (T, P, E) formulation:

(a) Predict whether a loan applicant will default, using ten years of labelled repayment records.

(b) Discover behavioural customer segments among 2 million customers for which no labels exist. (c) Estimate the next-month credit-card spend (in ₹) of each customer.

(d) Learn an ATM cash-refill policy that maximises cash availability minus transport cost through repeated trial and feedback.

(e) Only 2% of 500,000 transaction records are labelled fraud/genuine; the rest are unlabelled. Build a fraud detector that exploits both kinds of data.

Finally, justify which of the five systems is most harmed by label noise.

(5) (CO1) (BL4)

Task 2. A screening test for a disease with prevalence 2% has sensitivity 95% and specificity 90%. Calculate the probability that a person who tests positive actually has the disease, and analyse why the answer is much lower than the sensitivity.

(2) (CO1) (BL4)

Task 3. The joint probability mass function of a discrete random vector (X, Y) is given below.

X \ Y	Y = 0	Y = 1
X = 0	0.10	0.20
X = 1	0.15	0.25

Task 1: Design of learning systems

System	Paradigm	Task	Experience
Loan default	Supervised	classi-faction	10 years labelled data
Customer Segments	Unsupervised	Clustering	2M unlabled data
Next-month Spend	Supervised	Regression	past spending data
ATM refill	Reinforcement	control	Repeated trails
Fraud detection	Semi-Supervised	classific-ation	10,000 labelled

Task 2: Bayes Theorem

Given

prevelance = 0.02

Sensitivity = 0.95

False positive rate = 0.10

$$P(D/+) = 0.95(0.02) / 0.95(0.02) + 0.10(0.98)$$

$$= 0.019 / 0.117 = 0.162$$

It is lower than the Sensitivity because the disease is rare.

Task 3 : Joint Probability Distribution

a) Marginal probabilities

$$P(X) = (0.30, 0.40, 0.30)$$

$$P(Y) = (0.45, 0.55)$$

b) Mean & variance

$$E(X) = 0(0.3) + 1(0.4) + 2(0.3) =$$

$$E(X^2) = 1.6$$

$$\text{Var}(X) = 1.6 - 1^2 = 0.6 \quad E(Y) = 0.$$

c) Conditional probability

$$P(X=2 / Y=1) = 0.10 / 0.55 = 0.182$$

d) Covariance & correlation

$$E(XY) = 0.45$$

$$\text{Cov}(X,Y) = 0.45 - (1)(0.55) = -0.10$$

$$\rho = -0.10 / \sqrt{\sqrt{0.6}\sqrt{0.2475}} \approx -0.26$$

e) Independence

$$P(0,0) = 0.10 \neq P(X=0)\,P(Y=0).$$

Task 4 : Gaussian Distribution & Variance

Given $X \sim N(50, 16)$, so $\sigma = 4$

(a) probabilities

$$P(X > 58) = P(Z > 2)$$

$$= 1 - 0.9772 = 0.228$$

$$P(46 < X < 54) = P(-1 < Z < 1)$$

$$= 2(0.8413) - 1 = 0.6826$$

Task 3: Joint Probability Distribution

a) Marginal probabilities

P(x) = (0·30, 0·40, 0·30)

P(y) = 0·45, 0·55)

Variance

E(x) = 0(0·3) + 1(0·4) + 2(0·3) = 1

E(x2) = 1·6

= 1·6 − 1 = 0·6 E(y) = 0·55

ability

10/0·55 = 0·182

on

= −0·10

−0·26

ML & unbiased variance

Readings: 12, 15, 11, 14, 13

$\mu = 65/5 = 13$

sum of squared = 10 $\sigma^2_{ML} = 10/5 = 2$

$\sigma^2_{unbiased} = 10/4 = 2.5$

ML variance is biased because it divides by n instead.

Task 5: Information Theory

Probabilities : 1/2, 1/4, 1/8, 1/8

(a) Entropy $H = -\sum p_i \log_2 p_i$

$= 1/2(1) + 1/4(2) + 1/8(3) + 1/8(3)$

$= 1.75$ bits

(b) uniform source

$H = \log_2 4 = 2$ bits

(c) Redundancy $R = 2 - 1.75 = 0.25$ bits

(d) Avg code $L = 1/2(1) + 1/4(2) + 1/8(3) + 1/8(3)$

$= 1.75$ bits

the code is optimal $L = H$

Task 6 : KL Divergence

$P = (0.5, 0.25, 0.25)$ $Q = (1/3, 1/3, 1/3)$

(a) KL (P||Q) $D_{KL}(P||Q) = \sum P_i \log_2 \frac{P_i}{Q_i}$

$= 0.085$

(b) KL (Q||P) $D_{KL}(Q||P) = \sum Q_i \log_2 \frac{Q_i}{P_i}$

$= 0.082$ bits

(c) Asymmetry $D_{KL}(P||Q) \neq D_{KL}(Q||P)$

Task 7: Entropy and Mutual Informat

Given

0.4, 0.1, 0.1, 0.4

$H(A) = H(B) = 1$ bit

$H(A,B) = 1.722$ bits

$H(A/B) = H(A,B) - H(B)$

$= 0.722$ bits

$I(A;B) = H(A) - H(A/B)$

$= 0.278$ bits

Task 8: Bayesian classification

$\mu_1 = 2$ $\mu_2 = 6$ $\sigma = 2$

$P(w_1) = 0.6$ $P(w_2) = 0.4$

P = (0.5, 0.25, 0.25) Q = (1/3, 1/3, 1/3)

KL (P||Q) $D_{KL}(P||Q) = \sum P_i \log_2 \frac{P_i}{Q_i}$

= 0.085

Q||P) $D_{KL}(Q||P) = \sum Q_i \log_2 \frac{Q_i}{P_i}$

= 0.082 bits

Q||Q) ≠ DKL (Q||P)

Mutual Information

Decision Threshold

$$x^* = \frac{\mu_1 + \mu_2}{2} + \frac{\sigma^2}{\mu_2 - \mu_1} \ln \frac{P(w_i)}{P(w_2)}$$

$$= 4 + \ln(1.5) = 4.405$$

b) probability of error

$$P(\text{error}) = 0.6\,(0.1145) + 0.4\,(0.2127)$$

$$= 0.154$$

(c) Effect of prior

As $P(w_1)$ increases, the threshold moves right, expanding the decision region for w_1.

9 :- Min Risk classification

(a) Conditional risk

$$R(\alpha_1) = 0.3(0) + 0.7(8) = 5.6$$

$$R(\alpha_2) = 0.3(50) + 0.7(0) = 1.5$$

(b) Declare disease (α_1)

c) $P > 8/50 + 8 = 0.138$

Task 10 Bias Variance Trade off

Degree	Train MSE	Validation MSE	Gap
1	9	9.5	0.5
2	4.1	4.6	0.5
3	2.2	2.9	0.7
5	0.9	3.6	2.7
9	0.1	11.8	11.7

Task 11 Bayesian Estimation

(a) Posterior Beta(2+7, 2+3) = Beta(9, 5)

(b) for 10 tosses

Estimate	Value
ML	0.7
MAP	0.667
Posterior Value	0.643

c) For 100 tosses

	Value
ML	0.7
MAP	0.69
posterior mean	0.69

d) The prior pulls estimates towards 0.5

Task 12 ICU Sepsis prediction

a) T = Predict Sepsis b) P: Sensitivity

E = Historical ICU records

b) TP = 45, FN = 5, FP = 190, TN = 760

c) PPV = 45/235 = 19.15 NPV = 760/765 = 99.3

d) Total costs

Model = £1,450

Alert = £ 4750

Newer alert = 5,000

e) Threshold = 5/100+5 = 4.8%.